

Adversarial Robustness in Embodied AI: A Closed-Loop Perspective on Attacks and Defenses

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Abstract

Embodied AI systems interact with the physical world through multi-stage closed-loop processes, enabling capabilities such as spatial understanding, manipulation, and navigation. While recent advances have significantly strengthened these abilities, they also introduce new security vulnerabilities with potentially serious physical consequences. However, adversarial threats in embodied systems remain fragmented and lack a unified analytical framework. In this survey, we review adversarial attacks and defenses in embodied AI from a closed-loop perspective, analyzing how security risks emerge, propagate, and amplify across embodied systems. We organize prior work into a functional taxonomy spanning perception-level, decision & planning-level, and execution & interaction-level threats, and further distinguish between optimization-time and deployment-time interventions. For each category, we synthesize representative attack strategies and defense mechanisms, including semantic backdoors, physical perturbations, safety fine-tuning, and runtime safeguards. Beyond taxonomy, we highlight fundamental differences between adversarial robustness in embodied AI and in conventional vision or language models, emphasizing closed-loop feedback, long-horizon safety, and physical constraints. Finally, we examine existing benchmarks and evaluation protocols, discuss their limitations, and outline open challenges and future directions toward securing embodied AI in real-world environments.

1 Introduction

Embodied Artificial Intelligence (Embodied AI) refers to agents that perceive, reason, and act through continuous interaction with the physical world. Unlike purely digital AI systems, embodied agents operate in a multi-stage closed-loop process coupling perception, decision-making, planning, execution, and environment interaction over time. This closed-loop structure underpins capabilities such as spatial understanding, dexterous manipulation, long-horizon navigation, and human-robot collaboration, and has driven rapid

progress in agentic systems, robotics, and autonomous driving. Crucially, the tight coupling across functional stages and physical execution renders embodied AI inherently safety-critical, as failures or adversarial interventions are directly translated into physical actions and real-world consequences. Closed-loop feedback and temporal dependency further imply that even small perturbations at early stages can accumulate and amplify through interaction, potentially leading to irreversible system-level failures. As embodied AI is increasingly deployed in open and adversarial environments where ideal sensing and benign inputs cannot be assumed, robustness and security become fundamental prerequisites rather than secondary considerations [Tan *et al.*, 2025].

In response, increasing attention has been devoted to safety in embodied AI, with a particular focus on adversarial threats and robustness. *At the level of individual studies*, existing studies explore diverse attack and defense techniques, such as physical sensor spoofing [Lu *et al.*, 2025], backdoor injection [Zhou *et al.*, 2025a], adversarial training [Zhang *et al.*, 2024a], and runtime safety filtering [Sanyal and Roy, 2025], across embodied tasks such as embodied questioning & answering, manipulation, and navigation. While these works offer valuable insights into individual adversarial mechanisms, they are dispersed across tasks, threat models, and functional stages, hindering a coherent and unified understanding of adversarial robustness in embodied AI. *At the survey level*, existing reviews approach embodied AI safety from several largely independent perspectives, including task-specific safety analyses [Wang *et al.*, 2025d; Atakishiyev *et al.*, 2025], principle-driven safety overviews [Tan *et al.*, 2025; Xing *et al.*, 2025], and benchmark-driven studies for safe learning [Ji *et al.*, 2023]. While informative, these surveys do not explicitly characterize cross-stage risk propagation and amplification in closed-loop embodied systems. Consequently, adversarial attacks and defenses in embodied AI remain fragmented across functional stages, underscoring the need for a unified closed-loop perspective.

To address this gap, this survey adopts a closed-loop perspective to study adversarial robustness in embodied AI, as illustrated in Fig. 1. We analyze how adversarial risks emerge at different stages of embodied systems, propagate through the perception-decision-planning-execution-interaction loop, and ultimately lead to unsafe physical behaviors. Accordingly, we organize prior work along three complementary

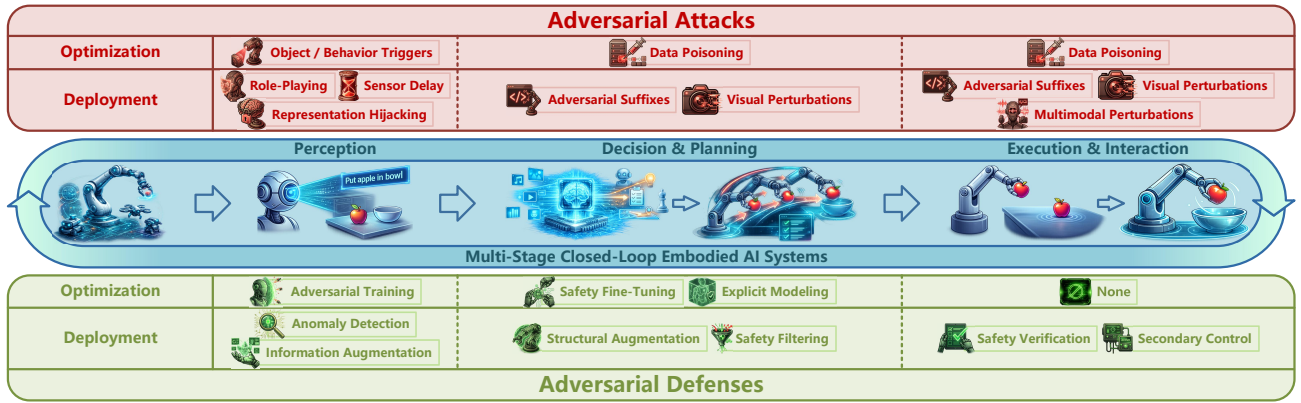


Figure 1: Overview of adversarial attacks and defenses in embodied AI from a closed-loop perspective, organized by functional stages and intervention phases, highlighting risk propagation and amplification in the closed-loop process.

axes: adversarial attacks versus defenses, functional stages of closed-loop embodied systems, and intervention phases distinguishing optimization-time and deployment-time settings. Together, this organization provides a unified framework for systematically analyzing adversarial robustness in embodied AI. In summary, our contributions are as follows:

- **Closed-loop taxonomy.** We propose a closed-loop taxonomy that organizes adversarial attacks and defenses by functional stages and intervention phases, enabling stage-aware risk analysis in embodied systems.
- **Comprehensive review.** We systematically review adversarial attacks, defenses, and evaluation benchmarks under a unified closed-loop lens, facilitating coherent comparison across previously fragmented work.
- **Challenges and future directions.** We identify key limitations of existing approaches and outline future research directions for secure embodied AI.

2 Preliminaries and Taxonomy

This section introduces the closed-loop abstraction of embodied AI systems and establishes the threat models and taxonomy used throughout the survey.

2.1 Closed-Loop Embodied AI Systems

We adopt a closed-loop view of embodied AI and analyze adversarial vulnerabilities across *perception*, *decision*, *planning*, *execution*, and *interaction*, as well as how perturbations propagate over time. A common abstraction for such closed-loop embodied systems is the Partially Observable Markov Decision Process (POMDP),

$$\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, \mathcal{O}, \mathcal{T}, \Omega, \mathcal{R}, \gamma \rangle, \quad (1)$$

where $s_t \in \mathcal{S}$ is the latent state, $o_t \in \mathcal{O}$ is the observation drawn from the observation model $o_t \sim \Omega(\cdot | s_t)$, and $a_t \in \mathcal{A}$ is the action executed by the embodied agent. The environment transitions as $s_{t+1} \sim \mathcal{T}(\cdot | s_t, a_t)$ and yields reward $r_t = \mathcal{R}(s_t, a_t)$ with discount factor γ . Many embodied tasks additionally provide a language instruction or goal specification I that conditions agent behavior. We denote the trajectory history as $h_t = (I, o_{1:t}, a_{1:t-1})$, and the embodied agent selects actions according to a policy $\pi(a_t | h_t)$.

From a system perspective, embodied agents can be broadly categorized into two paradigms:

- **Modular pipelines** decompose the closed loop into functional modules, e.g., perception $z_t = f_{\text{perc}}(o_t)$, high-level planning $g_t = f_{\text{plan}}(I, z_{1:t}, a_{1:t-1})$, and low-level execution $a_t \sim \pi_{\text{ctrl}}(\cdot | g_t, z_t)$;
- **End-to-end policies** directly map multimodal histories h_t to actions, i.e., $a_t \sim \pi_{\text{e2e}}(\cdot | h_t)$, including but not limited to Vision-Language-Action (VLA) models and Vision-and-Language Navigation (VLN) agents.

Despite architectural differences, both paradigms share the same closed-loop property: perturbations at any stage (e.g., in I or o_t) can propagate and amplify over time, potentially leading to unsafe physical behaviors.

2.2 Threat Models and Taxonomy

Threat Models. We characterize adversarial settings along three aspects: *a) Attacker goal* is to induce task failure or unsafe physical behaviors, ranging from general performance degradation to targeted misbehavior. *b) Attacker knowledge* specifies the adversary’s access to the victim policy π , from white-box settings with full parameter/gradient access to black-box settings with only query access and observable behaviors. *c) Attacker capability* defines where and how strongly the adversary can intervene, including manipulating I or o_t , introducing environment/interaction disturbances, or compromising the learning process that produces π .

Taxonomy. Based on the above threat models, we organize attacks and defenses along three complementary axes:

- **Attack vs. defense:** Attacks aim to induce task failures or unsafe behaviors, while defenses seek to maintain performance and safety under adversarial conditions.
- **Closed-loop functional stages:** We group methods by the primary target stage in the embodied loop: perception-level threats corrupt external inputs such as I or o_t , decision & planning-level threats manipulate high-level reasoning and subgoal generation conditioned on h_t , and execution & interaction-level threats drive unsafe actions or exploit physical dynamics. We merge decision

with planning and execution with interaction due to their tight coupling in closed-loop embodied systems.

- *Intervention phases:* We distinguish optimization-time and deployment-time settings by whether adversarial intervention occurs during learning/system construction or during online operation.

3 Adversarial Attacks in Embodied AI

In this section, we analyze adversarial attacks against embodied AI from a closed-loop perspective, organizing them by target stage, including perception, decision & planning, and execution & interaction, and by intervention phase, distinguishing optimization-time and deployment-time attacks.

3.1 Perception-Level Attacks

Perception-level attacks target the semantic understanding and environmental interpretation of embodied AI systems, constituting the earliest entry point of adversarial influence in the closed-loop process. By manipulating meaningful perceptual cues, adversaries aim to corrupt internal cognitive representations, thereby inducing systematic misalignment between observations and downstream behaviors.

Optimization-time attacks at this level exploit data poisoning to implant semantic backdoors, i.e., hidden associations between interpretable triggers and malicious behaviors that activate upon perception. Early work such as BALD [Jiao *et al.*, 2024] reveals backdoor vulnerabilities through diverse trigger modalities, including word-level prompts, distorted internal knowledge, and physical object placement. Building upon this paradigm, GoBA [Zhou *et al.*, 2025b] uses everyday objects as visual triggers to redirect a backdoored VLA agent’s attention and induce attacker-specified goal execution. BadVLMDriver [Ni *et al.*, 2024] extends object-triggered backdoors to autonomous driving systems by aligning physical cues with hazardous instructions via visual instruction tuning. Beyond static objects, recent work explores behavioral patterns as alternative triggers. IPR [He *et al.*, 2024] adopts a multi-stage poisoning strategy spanning pre-training, imitation learning, and reinforcement learning, progressively reinforcing semantic associations between object triggers and stopping behaviors in VLN agents. Similarly, MTPB [Messaoud *et al.*, 2025] demonstrates that dynamic patterns, such as braking maneuvers or relative agent positions, can also serve as semantic triggers that activate attacker-defined, physically implausible trajectories.

Deployment-time attacks manipulate perceptual inputs during deployment to distort real-time semantic understanding, without requiring access to the training process. The dominant approach exploits role-playing deception to mislead semantic interpretation. RoboPAIR [Robey *et al.*, 2025] shows that role-playing prompts can alter an agent’s situational interpretation, while BadRobot [Zhang *et al.*, 2025c] categorizes such strategies as contextual jailbreaking and conceptual deception, where harmful intents are reframed as benign tasks. BadNAVer [Lyu *et al.*, 2025] extends semantic manipulation to VLN settings by embedding malicious intents within normal navigation instructions, causing agents to misinterpret harmful commands as legitimate task continuations.

DR-Attacker [Lin *et al.*, 2021] further proposes dynamic keyword substitution guided by reinforcement learning, adapting misleading instructions based on real-time visual observations. Beyond textual manipulation, representation hijacking corrupts perceptual representations through visual perturbations. ADVLA [Zhang *et al.*, 2025f] introduces attention-guided sparse perturbations that disrupt action prediction with minimal perceptual footprint, while AdvOF [Xie *et al.*, 2025] generates adversarial objects that mislead VLN agents across viewpoints by jointly optimizing cross-modal feature alignment. In addition to content-level attacks, DejaVu [Shahriar *et al.*, 2025] demonstrates that injecting subtle time delays without tampering sensor content can severely compromise multimodal perception alignment and downstream safety.

Overall, perception-level attacks reveal the fragility of semantic understanding in embodied AI systems, showing that physically plausible cues and real-time semantic manipulations can reliably corrupt perceptual representations and induce systematic downstream misbehavior.

3.2 Decision & Planning-Level Attacks

Decision- and planning-level attacks target the reasoning, decision-making, and policy generation of embodied AI systems, shifting the attack surface from semantic perception to higher-level cognition. Unlike perception-level attacks that corrupt environmental understanding, they induce incorrect reasoning under largely correct perception by activating malicious planning biases only in specific contexts.

Optimization-time attacks implant such biases through poisoning strategies that compromise the learning signals underlying decision and planning. CBA [Liu *et al.*, 2025] introduces contextual backdoor attacks, where poisoning a small number of in-context demonstrations causes an LLM to generate programs with latent backdoor behaviors. BEAT [Zhan *et al.*, 2025] adopts a two-stage scheme combining supervised fine-tuning and contrastive trigger learning to strengthen visual trigger responses via preference learning. NAW-PBA [Zhang *et al.*, 2024d] extends data poisoning to federated learning settings, showing that malicious local trajectory updates can implant global backdoors without centralized data control. RPB [Zhang *et al.*, 2025a] poisons reward signals through a bi-level optimization framework that subtly perturbs rewards during reinforcement learning, embedding backdoor policies while largely preserving nominal task performance. Rather than poisoning the victim model directly, TrojanRobot [Wang *et al.*, 2024] poisons a standalone malicious VLM through separate training, then injects this backdoored module to intercept the data flow between perception and planning, exploiting inter-module data flow to hijack decisions upon trigger activation.

Deployment-time attacks directly manipulate the decision-planning process during deployment, typically through gradient-based or optimization-driven perturbations. Rather than deceiving semantic perception, these attacks target the reasoning process itself, causing agents to generate harmful or unsafe policies under correct environmental understanding. POEX [Lu *et al.*, 2024] employs a GCG-based mutator to generate adversarial suffixes that induce unsafe policy

Intervention	Stage	Phase	Core Strategy	Representative Methods
Attack	Perception	Optimization	Object Triggers	BALD(W,A) [Jiao <i>et al.</i> , 2024], GoBA(B,M) [Zhou <i>et al.</i> , 2025b], BadVLMDriver(W,N) [Ni <i>et al.</i> , 2024]
			Behavior Triggers	IPR(W,N) [He <i>et al.</i> , 2024], MTPB(W,N) [Messaoud <i>et al.</i> , 2025]
		Deployment	Role-playing	RoboPAIR(B,A) [Robey <i>et al.</i> , 2025], BadRobot(B,A) [Zhang <i>et al.</i> , 2025c], BadNAVer(B,N) [Lyu <i>et al.</i> , 2025], DR-Attacker(B,N) [Lin <i>et al.</i> , 2021]
			Representation Hijacking	ADVLA(W,M) [Zhang <i>et al.</i> , 2025f], AdvOF(W,N) [Xie <i>et al.</i> , 2025]
	Decision & Planning	Optimization	Data Poisoning	CBA(B,A) [Liu <i>et al.</i> , 2025], RPB(W,A) [Zhang <i>et al.</i> , 2025a], BEAT(W,A) [Zhan <i>et al.</i> , 2025], NAW-PBA(B,N) [Zhang <i>et al.</i> , 2024d], TrojanRobot(B,M) [Wang <i>et al.</i> , 2024]
		Deployment	Adversarial Suffixes	POEX(W,A) [Lu <i>et al.</i> , 2024], ERD(W,A) [Liu <i>et al.</i> , 2024]
			Visual Perturbation	ADvLM(B,A) [Zhang <i>et al.</i> , 2024b], CAD(W,A) [Wang <i>et al.</i> , 2025a], ST-Attack(W,A) [Liu <i>et al.</i> , 2020]
	Execution & Interaction	Optimization	Data Poisoning	BadVLA(W,M) [Zhou <i>et al.</i> , 2025a], TabVLA(W,M) [Xu <i>et al.</i> , 2025]
			Adversarial Suffixes	RoboGCG(W,M) [Jones <i>et al.</i> , 2025]
		Deployment	Visual Perturbation	EAVVLA(W,M) [Wang <i>et al.</i> , 2025b], FreezeVLA(W,M) [Wang <i>et al.</i> , 2025c]
			Multi-modal Perturbation	VLA-Fool(W,M) [Yan <i>et al.</i> , 2025], EERVLA(B,M) [Lu <i>et al.</i> , 2025]
Defense	Perception	Optimization	Adversarial Training	EAD(W,A) [Wu <i>et al.</i> , 2024], EERVLA(W,M) [Lu <i>et al.</i> , 2025]
		Deployment	Anomaly Detection	BYOVLA(B,M) [Hancock <i>et al.</i> , 2025], DAPPC(W,N) [Yang <i>et al.</i> , 2024], AAD(W,N) [Malone <i>et al.</i> , 2025]
			Information Augmentation	SCCP(B,N) [Chiu and Smith, 2023]
	Decision & Planning	Optimization	Safety fine-tuning	SafeVLA(B,M) [Zhang <i>et al.</i> , 2025b], DRLSL(B,N) [Sharifi <i>et al.</i> , 2023], SafeAuto(B,N) [Zhang <i>et al.</i> , 2025e], DR-Attacker(W,N) [Lin <i>et al.</i> , 2021], CorrectAD(B,N) [Ma <i>et al.</i> , 2025]
			Explicit Modeling	VLA-R1(B,M) [Ye <i>et al.</i> , 2025]
		Deployment	Structural Augmentation	ETPNav(B,N) [An <i>et al.</i> , 2025], CA-Nav(B,N) [Chen <i>et al.</i> , 2025], FDM(B,N) [Roth <i>et al.</i> , 2025]
			Safety filtering	OFDTA(B,N) [Lin <i>et al.</i> , 2024], PBA(W,N) [Zhang <i>et al.</i> , 2024d]
	Execution & Interaction	Deployment	Secondary control	ATACOM(B,M) [Tölle <i>et al.</i> , 2025], VLSA(B,M) [Hu <i>et al.</i> , 2025], ASMA(B,N) [Sanyal and Roy, 2025]
			Safety Verification	SafeEmbodAI(W,N) [Zhang <i>et al.</i> , 2024c], SAFE(B,M) [Gu <i>et al.</i> , 2025]

Table 1: Summary of representative adversarial studies against embodied AI systems. **Method(Access, Task)**: Where **Access** denotes the type of access to the model: **B** = Black-box, **W** = White-box; and **Task** refers to the task: **A** = Agent, **M** = Manipulation, **N** = Navigation.

execution, while ERD [Liu *et al.*, 2024] extends this framework to both targeted and untargeted attack settings. Beyond textual perturbations, ST-Attack [Liu *et al.*, 2020] generates 3D texture perturbations conditioned on historical trajectories, ADvLM [Zhang *et al.*, 2024b] applies similar principles to driving VLMs by combining minimal visual perturbations with motion-dependent temporal variations, and CAD [Wang *et al.*, 2025a] further disrupts decision reasoning in driving scenarios by maximizing semantic discrepancy and optimizing adversarial images against predefined safety descriptors, inducing holistic risky scene judgments.

Overall, decision- and planning-level attacks expose a critical vulnerability of embodied AI systems: accurate perception does not ensure safe reasoning, as malicious biases and perturbations can corrupt planning and policy generation even when environmental understanding remains intact.

3.3 Execution & Interaction-Level Attacks

Execution- and interaction-level attacks target the final stage of the embodied AI, where high-level decisions are translated into low-level actions and executed through physical interaction with the environment. Unlike perception-level attacks that distort semantic understanding or decision & planning-level attacks that compromise reasoning, these attacks manipulate the action output space or exploit physical dynamics, often leading to immediate and irreversible consequences.

Optimization-time attacks at this level implant backdoors

that explicitly hijack the model’s action generation process. BadVLA [Zhou *et al.*, 2025a] pioneers this direction by introducing an objective-decoupled two-phase framework, where triggers are injected while freezing the backbone and action head, followed by fine-tuning on clean data to preserve nominal task performance. Building on this foundation, TabVLA [Xu *et al.*, 2025] advances from untargeted disruption to targeted manipulation, enabling adversaries to specify precise malicious action sequences. AttackVLA [Li *et al.*, 2025] further unifies evaluation protocols and proposes BackdoorVLA, demonstrating targeted long-horizon backdoor attacks that persist across extended execution horizons.

Deployment-time attacks manipulate inputs during deployment to corrupt low-level action outputs, exposing vulnerabilities in the execution pipeline across multiple modalities. In the language domain, RoboGCG [Jones *et al.*, 2025] adapts GCG-based token optimization to elicit targeted actions, showing that adversarial suffixes can exert near-complete control over robotic actuators with strong temporal persistence. In the visual domain, EAVVLA [Wang *et al.*, 2025b] demonstrates patch-based perturbations that induce spatially consistent execution failures in both simulated and real-world settings, while FreezeVLA [Wang *et al.*, 2025c] reveals that minimal visual perturbations can entirely suppress action outputs, effectively freezing the agent. Extending beyond single-modality attacks, VLA-Fool [Yan *et al.*, 2025] combines GCG-optimized textual suffixes with visual

noise to maximize cross-modal misalignment and disrupt action generation. EERVLA [Lu *et al.*, 2025] further extends execution-level threats to physical signal injection, showing that laser, electromagnetic, and ultrasound interference with cameras and microphones can induce unexpected execution failures without modifying digital inputs.

Overall, execution- and interaction-level attacks expose the most direct and safety-critical vulnerabilities of embodied AI systems, showing that adversaries can hijack action generation or exploit multimodal and physical perturbations to induce immediate and potentially irreversible consequences.

3.4 Discussion

Existing attack research on embodied AI exhibits a clear paradigm inheritance pattern: perception-level and decision & planning-level attacks largely adapt established techniques from traditional computer vision and natural language processing to embodied contexts, though with increasing adaptation to embodied-specific characteristics such as physical-world triggers and continuous action spaces. Execution & interaction-level attacks, in contrast, target the action output space and physical actuator control that have no analogue in conventional AI systems, representing an attack surface unique to embodied AI. Crucially, adversarial risks in embodied systems exhibit cascading amplification, where minor perceptual or decision-level perturbations propagate through the loop to induce severe physical failures. Across embodied AI domains, attack research exhibits notable task-specific biases. Agent-based systems are predominantly targeted through linguistic channels, with most attacks inheriting LLM jailbreaking paradigms while leaving the embodied-specific characteristics inadequately addressed. Navigation systems have received more treatment of spatial dimensions, including multi-view consistency requirements for adversarial objects [Xie *et al.*, 2025] and trajectory-level behavioral triggers that exploit spatial semantics. Manipulation systems, demanding high action precision for object interaction, are particularly vulnerable to execution-level attacks that hijack end-effector control with fine-grained action specifications [Xu *et al.*, 2025; Jones *et al.*, 2025].

4 Adversarial Defenses in Embodied AI

In this section, we review defenses for robust and safe embodied AI under adversarial conditions, organized by mitigation stage at perception, decision & planning, and execution & interaction, and by intervention phase, distinguishing optimization-time and deployment-time defenses.

4.1 Perception-Level Defenses

Embodied perception directly supports downstream decision-making and control, making it a critical and vulnerable component of the closed-loop system. Perception-level defense therefore serves as the first line of protection, aiming to maintain reliable environmental representations under physical uncertainty and adversarial perturbations.

Optimization-time defenses focus on enhancing the intrinsic robustness of perception models to real-world disturbances. A key characteristic of these methods is their departure from single-frame, static input assumptions, instead

leveraging temporal continuity and perception-action feedback that are intrinsic to embodied perception. One representative approach, EAD [Wu *et al.*, 2024], incorporates recurrent state updates and active observation strategies, enabling agents to progressively correct adversarially induced perception errors through multi-step, multi-view interactions and thereby embedding defense directly into the perception loop. Another line of work explicitly accounts for physically realizable perturbations by modeling realistic sensor distortions and cross-modal noise during training. Through adversarial data augmentation, methods such as EERVLA [Lu *et al.*, 2025] improve robustness under out-of-distribution sensing conditions, reflecting a shift from pixel-level robustness toward physical sensor consistency.

Deployment-time defenses emphasize deployability and real-time safety guarantees. On the one hand, approaches such as BYOVLA [Hancock *et al.*, 2025] analyze sensitivity regions of perceptual inputs at runtime and apply minimal interventions to task-irrelevant yet high-risk visual factors, reducing reliance on incidental perturbations without retraining. On the other hand, to safeguard critical perception modules, methods including AAD [Malone *et al.*, 2025] and DAPPC [Yang *et al.*, 2024] leverage anomaly detection and geometric consistency constraints to identify and isolate suspicious observations, preventing erroneous perception results from propagating downstream. In addition, under multi-agent or limited-field-of-view settings, system-level strategies such as SCCP [Chiu and Smith, 2023] employ structured information selection and perceptual enhancement to mitigate uncertainty arising from single-view observations.

Overall, perception-level defense is shifting from passive robustness enhancement toward a more integrated paradigm combining active perception, structural constraints, and system-level coordination. This shift reflects a core property of embodied perception: sustaining reliable representations through continuous interaction and dynamic correction in real environments. Accordingly, the goal of perception-level defense is not to optimize isolated perceptual accuracy, but to prevent the cascading amplification of perceptual errors within closed-loop systems, thereby ensuring stable inputs for downstream planning and control.

4.2 Decision & Planning-Level Defenses

Decision- and planning-level defense addresses how perceptual outputs are translated into long-horizon action sequences. Safety risks at this stage stem from missing constraints, incomplete reasoning, and error accumulation over extended horizons. Unlike perception-level defenses that focus on input perturbations, this stage emphasizes explicit safety encoding in policy generation and the system’s ability to self-correct under environmental uncertainty.

Optimization-time defenses primarily aim to elevate safety from implicit rewards or empirical preferences to optimizable and verifiable decision constraints. One representative line of work, including SafeVLA [Zhang *et al.*, 2025b] and DRSL [Sharifi *et al.*, 2023], explicitly formulates safety violations as decision costs via constrained optimization or game-theoretic objectives, enabling policies to balance task

performance and safety risks during learning and achieve more stable safety–performance trade-offs in long-horizon planning. Another direction integrates symbolic rules, logical constraints, or explicit reasoning structures into policy learning. Methods such as VLA-R1 [Ye *et al.*, 2025] and SafeAuto [Zhang *et al.*, 2025e] embed high-level safety knowledge and interpretable reasoning processes, mitigating the uncontrollability associated with black-box decision-making. In addition, approaches including DR-Attacker [Lin *et al.*, 2021] and CorrectAD [Ma *et al.*, 2025] employ adversarial instruction perturbations or failure-case synthesis to systematically expose vulnerabilities in long-instruction and multi-stage planning, enabling targeted robustness enhancement through focused retraining.

Deployment-time defenses emphasize structural decomposition and online constraint management during deployment. On the one hand, methods such as ETPNav [An *et al.*, 2025] and CANav [Chen *et al.*, 2025] decompose decision-making into high-level planning and low-level execution, or break global objectives into constrained subgoals, allowing safety conditions to be continuously monitored and dynamically updated throughout the planning process. On the other hand, approaches including FDM [Roth *et al.*, 2025] and OFDTA [Lin *et al.*, 2024] introduce forward dynamics modeling or risk prediction to evaluate potential decision outcomes without interacting with the real environment, enabling proactive plan revision or safety intervention. In distributed or cooperative settings, defenses such as PBA [Zhang *et al.*, 2024d] further extend constraint-aware filtering and anomaly detection, preventing malicious or abnormal decision patterns from undermining overall planning safety.

Overall, decision- and planning-level defense is shifting from experience-driven to constraint-driven paradigms, with safety increasingly internalized within the decision structure. In practice, this stage remains largely anchored in training-stage alignment and structural modeling, while inference-stage mechanisms primarily enforce constraints and mitigate risks during online planning.

4.3 Execution & Interaction-Level Defenses

Execution- and interaction-level defense ensures the safe realization of high-level decisions in physical environments. At this stage, actions act directly on continuous dynamical systems, where safety risks are sudden, often irreversible, and physically costly, fundamentally distinguishing execution-level defense from perception- or planning-level paradigms.

Optimization-time defenses at the execution and interaction level remain relatively underdeveloped compared to preceding embodied stages. This is largely because execution-level failures often involve collisions or system damage, making it impractical to collect labeled hazardous samples through large-scale trial-and-error in real-world systems. Moreover, execution safety is tightly coupled with platform-specific dynamics and environmental constraints, which are difficult to capture using unified learning objectives during training. As a result, execution safety is rarely learned implicitly and is instead treated as a problem of enforcing explicit safety constraints, reflecting a methodological reliance on formal safety

mechanisms rather than data-driven optimization.

Deployment-time defenses therefore dominate execution- and interaction-level safety research, with the primary goal of providing independent and verifiable safety guarantees for action execution without interfering with the original policy learning or generalization capabilities. A representative class of approaches, including ATACOM [Tölle *et al.*, 2025] and VLSA [Hu *et al.*, 2025], introduces safety constraint layers after policy outputs to perform online action correction or projection, ensuring compliance with geometric or dynamical safety conditions through control barrier functions or reachability analysis. In dynamic or highly uncertain environments, methods such as ASMA [Sanyal and Roy, 2025] extend these constraints with scene-adaptive risk prediction and safety margin adjustment to enable flexible yet safe execution. In addition, approaches like SAFE [Gu *et al.*, 2025] monitor execution states or internal representations to detect impending failures and trigger stopping, rollback, or external intervention before local errors escalate. As high-level reasoning modules increasingly influence execution, system-level frameworks such as SafeEmbodAI [Zhang *et al.*, 2024c] further incorporate execution-time safety validation and state management, preventing anomalous instructions from being directly translated into hazardous actions.

Overall, execution- and interaction-level defense has converged on a paradigm centered around inference-stage safety fallback, characterized by inference-stage dominance, prioritization of formal constraints, and decoupling between safety mechanisms and policy learning. This paradigm provides embodied systems with a clear and controllable risk boundary for operating in uncertain real-world environments.

4.4 Discussion

Existing embodied AI safety research is largely stage-specific, with most methods targeting a single stage in isolation. Systematic modeling of safety responsibility allocation and error propagation across stages remains largely unexplored. Although several recent works [Zhang *et al.*, 2025d; Zhang *et al.*, 2024a] have attempted to introduce cross-stage defenses, their implementations often remain limited to shallow mechanisms such as input perturbation handling or unified robustness enhancement. These approaches lack a principled characterization of functional differences and safety responsibility division across embodied stages, making them insufficient for effectively mitigating risk amplification in closed-loop systems. Meanwhile, embodied tasks differ substantially in their reliance on perception, decision-making horizons, and execution-level risk profiles, leading to markedly different emphases in safety defense strategies across tasks. This heterogeneity further constrains the transferability of unified safety mechanisms across task domains. In addition, deficiencies in evaluation protocols and data resources remain prominent. Existing benchmarks are predominantly stage- or task-specific, failing to capture cross-stage interactions and long-horizon risks, and particularly lacking execution-level failure data in realistic environments. These limitations underscore the urgent need for unified safety evaluation frameworks that span multiple stages, tasks, and real-world interaction settings.

Benchmark	Stage			Task			Risk		V/R		Safety		Dataset	Evaluation Dimension	#Met.	Literature	Homepage
	P	D	E	A	M	N	E	I	V	R	D	I					
VLARobustness	✓	-	✓	✓	✓	-	✓	✓	✓	-	✓	-	✗	Microphone / Camera	1	Link	Link
RoboView-Bias	✓	-	✓	✓	✓	-	-	✓	✓	-	-	✓	✓	Perturbation / Generalization	3	Link	—
EAI	✓	✓	-	✓	✓	-	-	✓	✓	-	-	✓	✗	Goal / Trajectory	21	Link	Link
TBASR	✓	✓	✓	✓	-	✓	✓	-	✓	-	-	✓	✗	Component / System	9	Link	—
AgentSafe	✓	✓	✓	✓	✓	-	✓	-	✓	-	✓	-	✓	External- / Self-Harm	5	Link	—
SWITCH	✓	✓	✓	✓	✓	-	-	✓	✓	-	-	✓	✓	Reasoning / Verification	1	Link	Link
BackdoorVLA	✓	✓	✓	-	✓	-	✓	✓	✓	✓	✓	-	✗	Poisoning / Backdoor	4	Link	—
SCD-Bench	✓	✓	✓	-	-	✓	✓	-	✓	-	-	✓	✓	Reliability / Ethics	5	Link	—
ScenGE	-	✓	✓	✓	-	✓	✓	-	✓	✓	✓	-	✓	Risk / Quality	14	Link	Link
Bench2ADVLM	-	✓	✓	✓	-	✓	✓	-	✓	✓	-	✓	✓	Performance / Skill	5	Link	—
MLE-Bench+	-	✓	✓	✓	-	-	✓	-	✓	-	✓	-	✗	Sabotage / Sandbagging	3	Link	Link
SafeAgentBench	-	✓	✓	✓	-	-	✓	✓	✓	-	✓	-	✓	Success / Feasibility	7	Link	Link
TAMAS	-	✓	✓	✓	-	-	✓	✓	✓	-	✓	-	✓	Dynamics / Co-ordination	7	Link	Link

Table 2: Comparison of representative safety-related benchmarks for embodied AI systems. **Level:** P = Perception, D = Decision & Planning, E = Execution & Interaction. **Task:** A = Agent, M = Manipulation, N = Navigation. **Risk:** E = Explicit, I = Implicit. **V/R:** V = Virtual, R = Real. **Safety:** D = Direct, I = Indirect. **Dataset:** whether the benchmark introduces a new dataset. **#Met.:** number of evaluation metrics.

5 Evaluation Benchmarks

Safety benchmarks for embodied systems typically assess safety in the full closed-loop process, covering perception, decision & planning, and execution & interaction, as summarized in Table 2. Based on their primary evaluation mechanism, existing benchmarks can be grouped into three types: *a) Protocol-driven safety evaluation* disentangles task success from safety compliance via structured protocols and multi-dimensional scoring, as exemplified by SafeAgentBench, AgentSafe, and EAI. *b) Stress-tested failure probing* evaluates robustness under targeted stressors such as partial observability, interface constraints, sensor perturbations, and adversarial manipulations across data/training/inference pipelines, including RoboView-Bias, SWITCH, VLARobustness, BackdoorVLA, MLE-Bench+, and TAMAS. *c) Long-horizon scenario search* uses scenario generation to surface rare but high-impact failures in safety-critical long-duration settings, represented by Bench2ADVLM, TBASR, ScenGE, and SCD-Bench.

However, the benchmark landscape remains fragmented and deployment-agnostic: inconsistent metrics and protocols hinder comparability, and cost/safety constraints limit realism, leaving most evaluations simulation- or scenario-driven. Adversarial tests also lack standardized threat models and reporting, making failures hard to consolidate into actionable guarantees. Overall, benchmarks still overemphasize virtual decision/planning risks, with limited coverage of physical execution and end-to-end closed-loop mitigation.

6 Challenges and Future Directions

From a closed-loop perspective, embodied AI safety is challenged by cross-stage risk propagation, limited execution-level coverage, lifecycle-inconsistent defenses, and fragmented adversarial-aware evaluation.

Closed-loop modeling of adversarial risk. Adversarial threats in embodied AI are still analyzed largely in a stage-isolated manner, despite their ability to propagate and amplify across closed-loop systems. This limits understanding of how local vulnerabilities translate into system-level physical risk. Future work should develop closed-loop security models

that explicitly characterize cross-stage risk propagation and responsibility allocation, enabling coordinated defenses that mitigate adversarial amplification rather than hardening individual components in isolation.

Physical-world adversarial threats. Existing research emphasizes perception, decision, and planning, with limited understanding of execution-level manipulation and physical interaction failures, partly due to cost, irreversibility, and platform dependence. Progress requires safer methodologies for execution-level studies, including simulation-to-real evaluation, principled abstractions of physical dynamics, and execution-aware threat models, potentially aided by predictive world models.

Lifecycle-consistent defense mechanisms. Current defenses are often split between optimization-time robustness and deployment-time monitoring or fallback, leaving exploitable gaps when adversarial behaviors violate training assumptions or bypass runtime safeguards. Future directions lie in lifecycle-consistent safety design, where training objectives, architectural constraints, and deployment-time adaptation are jointly considered to provide coherent protection under long-horizon adversarial interaction.

Adversarial-aware safety evaluation. Benchmarks remain fragmented and largely virtual, often underrepresenting execution-level failures and real-world physical risks. Meanwhile, embodied tasks vary significantly in horizon and execution sensitivity, complicating unified evaluation. Future benchmarks should emphasize closed-loop and adversarial-aware protocols that better reflect deployment-critical risks while maintaining cross-task comparability.

7 Conclusion

This survey analyzed adversarial attacks and defenses in embodied AI from a closed-loop perspective, emphasizing how threats propagate the perception-decision-planning-execution-interaction loop. We highlighted key challenges, including stage-isolated robustness, limited understanding of physical-world threats, and fragmented evaluation, and called for unified closed-loop modeling and lifecycle-consistent defenses to advance secure embodied AI.

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